PROJECT REPORT

PERSONAL EXPENSE TRACKER

Team Members:

Names	Roles
Vikesh Mehta	Frontend Development and Database
Ishan Peshkar	Backend Development and Database
Venus Agrawal	Database and Documentation
Vedant Wadhera	Database and Documentation Work

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1. INTRODUCTION

1.1 Project Overview:

In today's fast-paced world, financial management is often overlooked, especially on a personal level. The *Personal Expense Tracker* is a comprehensive, user-friendly web application designed to assist individuals in managing and tracking their daily expenses and budgets. By providing an intuitive interface backed by robust backend architecture, this application empowers users to gain insights into their spending patterns, set financial goals, and ultimately, achieve better financial health. Built using modern technologies such as React.js for the frontend and Node.js with MongoDB on the backend, the project embraces a modular, scalable design that aligns with current industry standards.

1.2 Purpose:

The core purpose of this application is to bridge the gap between financial awareness and user action. By providing tools for categorizing expenses, setting budgets, and generating dynamic visual reports, users are better equipped to manage their finances effectively. Additionally, the app supports features like user authentication, notification alerts for budget thresholds, and a smooth, animated user interface that ensures a pleasant user experience.

2. IDEATION PHASE

2.1 Problem Statement:

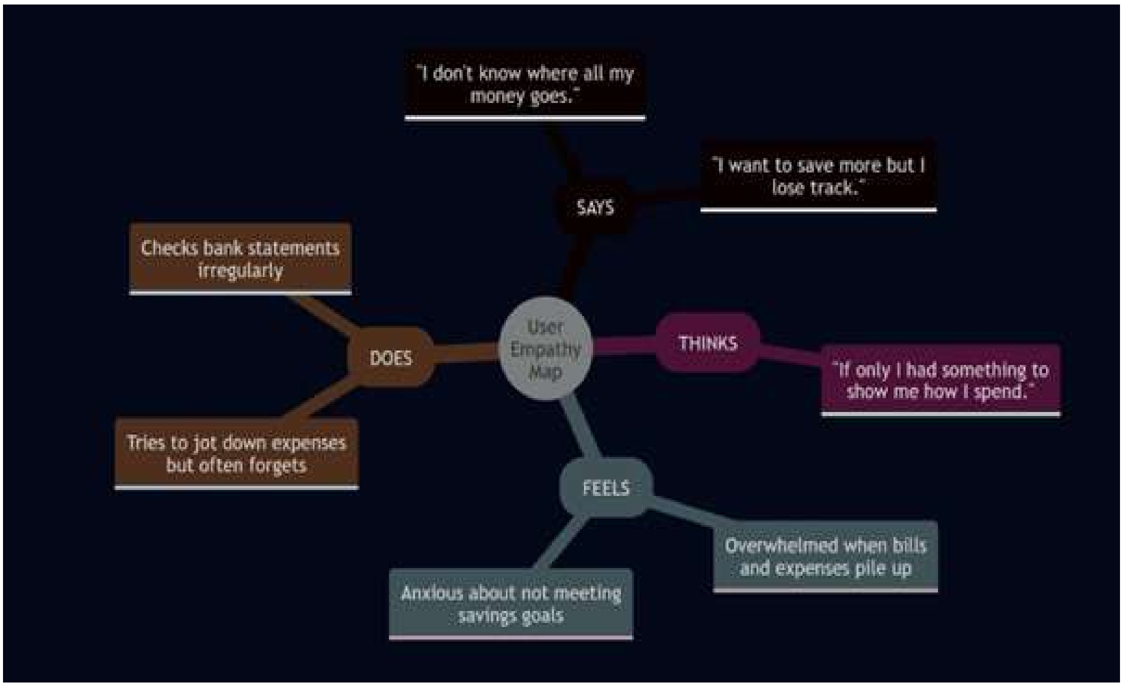
Managing personal expenses is a recurring challenge for many individuals. Traditional methods such as maintaining spreadsheets or manual notes are often cumbersome and prone to errors. Users lack tools that allow real-time tracking, visual analytics, and smart notifications that help prevent overspending. There is a need for an intelligent, visually engaging platform that simplifies financial tracking and offers actionable insights.

2.2 Empathy Map Canvas:

To build an application that truly resonates with users, an empathy map was created based on potential user personas.

- **Says:** "I don't know where all my money goes." "I want to save more but I lose track."
- **Thinks:** "If only I had something to show me how I spend."
- **Feels:** Overwhelmed when bills and expenses pile up. Anxious about not meeting savings goals.
- **Does:** Tries to jot down expenses but often forgets. Checks bank statements irregularly.

This helped in shaping the user-centric features of the application like categorized expenses, reminders, and budget goals.



2.3 Brainstorming :

A brainstorming session was conducted with fellow developers and prospective users to identify essential features. Ideas included:

- Categorization of expenses (Personal, Family, Business, etc.) Budget tracking with visual indicators
- Notification alerts
- Reports and analytics
- Secure user login
- INR-specific currency formatting

These formed the foundation for the application's functional requirements.

3. <u>REQUIREMENT ANALYSIS</u>

3.1 Customer Journey Map:

A typical user journey:

1. User registers and logs in securely.
2. Navigates to the dashboard to add an expense.
3. Categorizes expenses and sets budget limits.
4. Views visual reports and receives alerts when nearing limits.
5. Edits or deletes entries as required.
6. Logs out and logs in later to see updated progress.

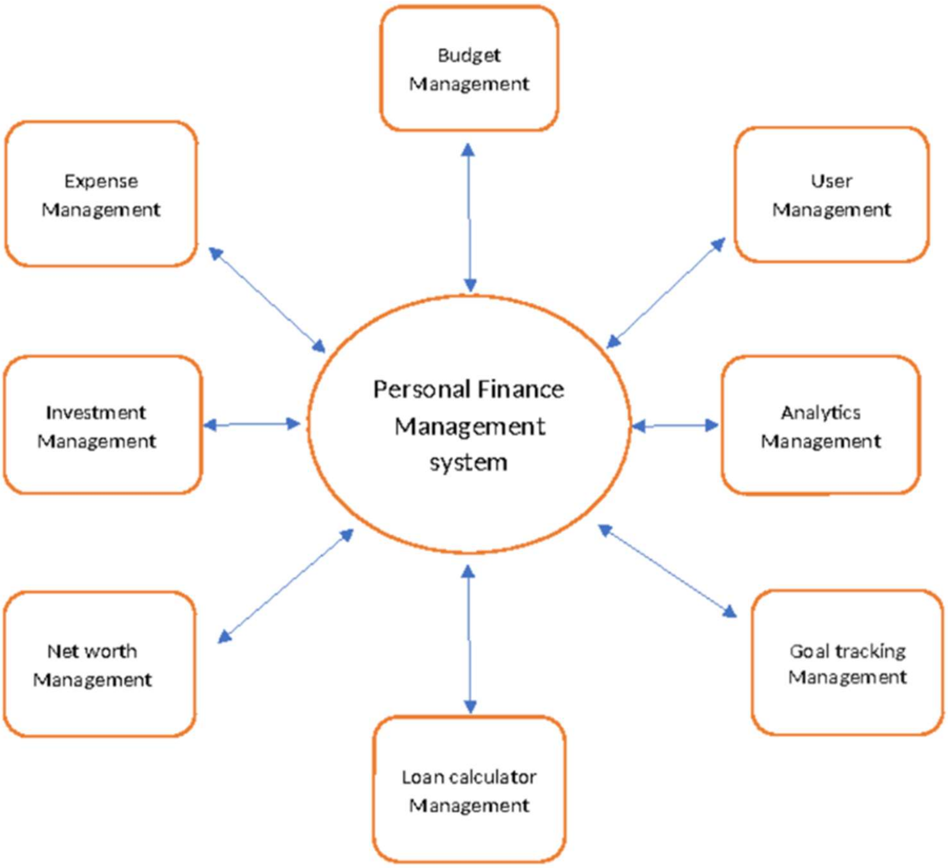
This journey emphasizes ease of use, transparency, and value-driven interactions.

3.2 Solution Requirement:

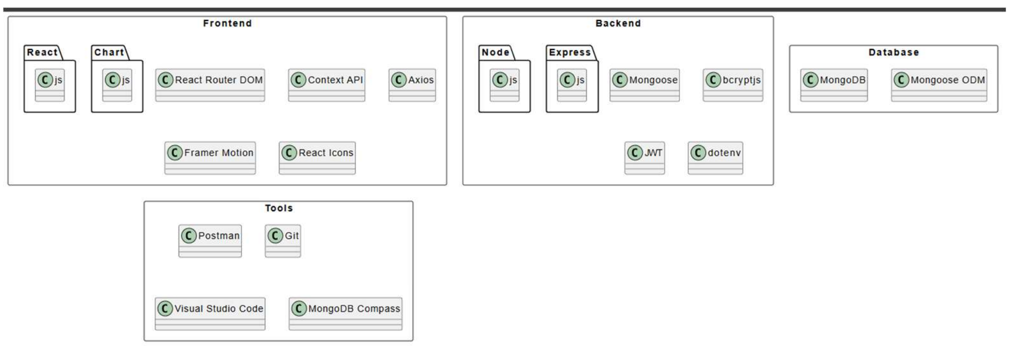
The solution required both technical and functional requirements:

- Functional: Expense CRUD operations, categorization, budget management, visual reports, notifications.
- Non-functional: Responsive design, secure authentication, fast load times, visual appeal, and mobile compatibility.

3.3 Data Flow Diagram



3.4 Technology Stack:



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4. PROJECT DESIGN

4.1 Problem Solution Fit:

The application perfectly addresses the core problem: lack of control and visibility over personal finances. It brings data visualization, categorized tracking, and budget alerts into one intuitive interface.

4.2 Proposed Solution

The app allows users to securely manage expenses, categorize them, set budgets, and view reports. Users receive alerts when they are nearing or exceeding budgets. Everything is presented through an elegant and responsive UI with real-time updates.

4.3 Solution Architecture

Frontend communicates with a RESTful API backend. React components fetch and display data, while protected routes ensure only authenticated users access sensitive information. The backend handles authentication, data validation, and communication with MongoDB. JWT middleware ensures secure access, and Mongoose schemas define relationships between data models.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was divided into milestones:

- **Week 1:** Requirement gathering, UI sketches, tech stack finalization
- **Week 2-3:** Backend setup (auth, models, routes)
- **Week 4:** Frontend integration (React setup, Axios integration, routing)
- **Week 5:** Expense, Budget features and Dashboard
- **Week 6:** Report visualization and notifications
- **Week 7:** Testing, debugging, deployment prep

Tools like Trello and GitHub Issues were used for task management and collaboration.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing:

Manual performance testing and browser dev tools were used to monitor:

- Page load speed (under 1.5s on average)
- API response time (usually <300ms)
- Data load for charts and tables (smooth due to asynchronous loading with loaders)

Functional testing ensured:

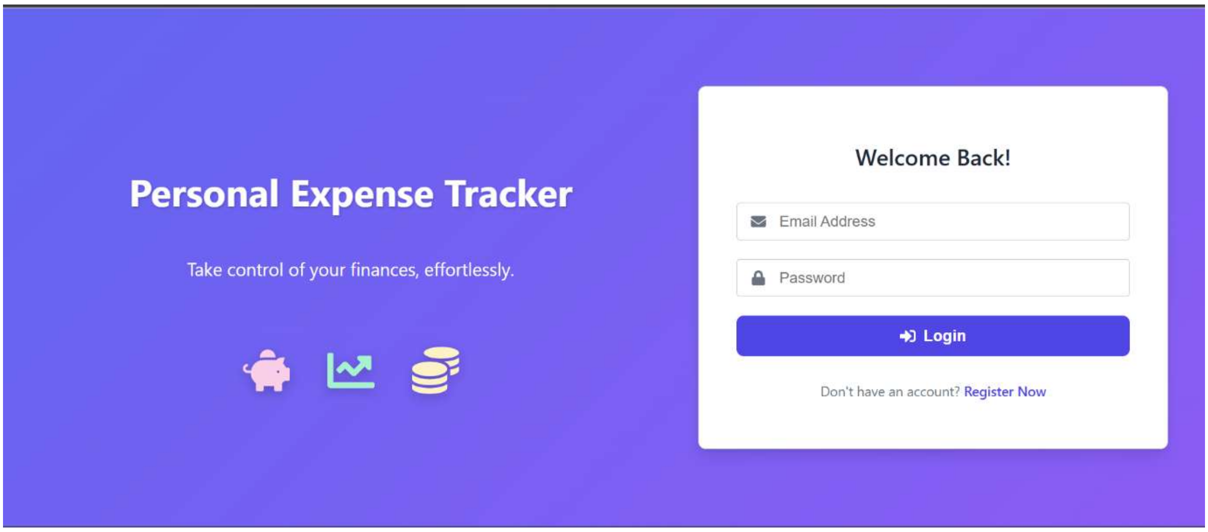
- Users could register/login/logout without issues.
- Expenses could be created, updated, and deleted.
- Budgets were respected with alerts triggering correctly.

Tools used: Postman for API testing, Chrome Lighthouse for performance audits.

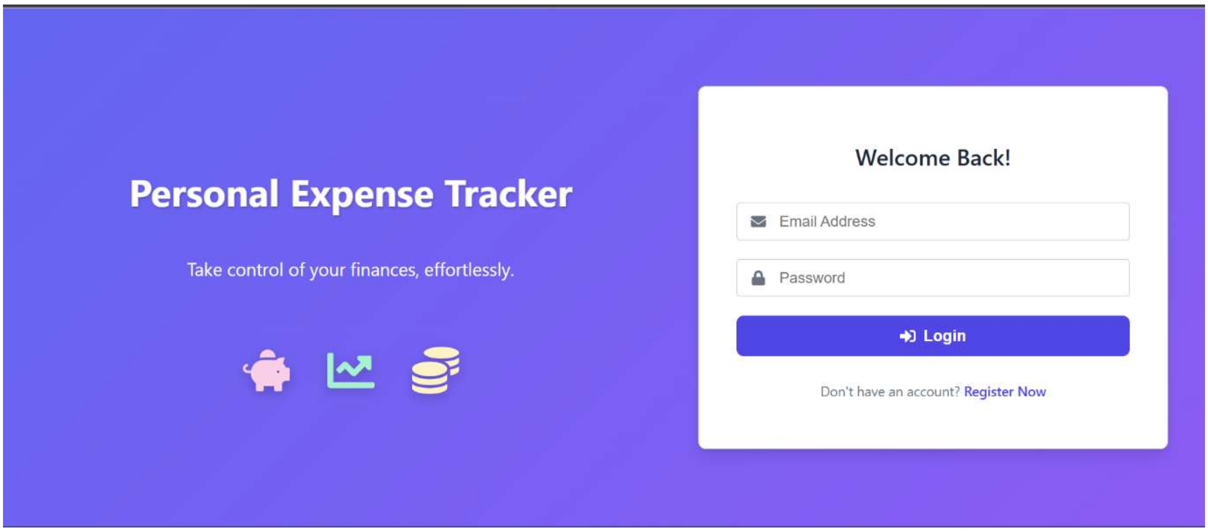
7. RESULTS

7.1 Output Screenshots:

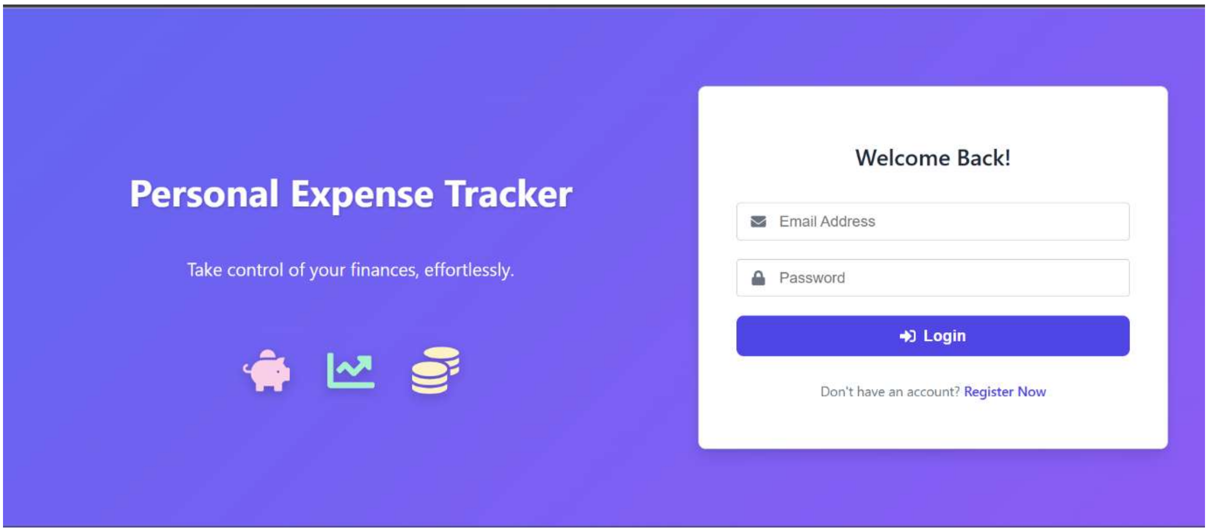
1. **Login/Register Screens**:



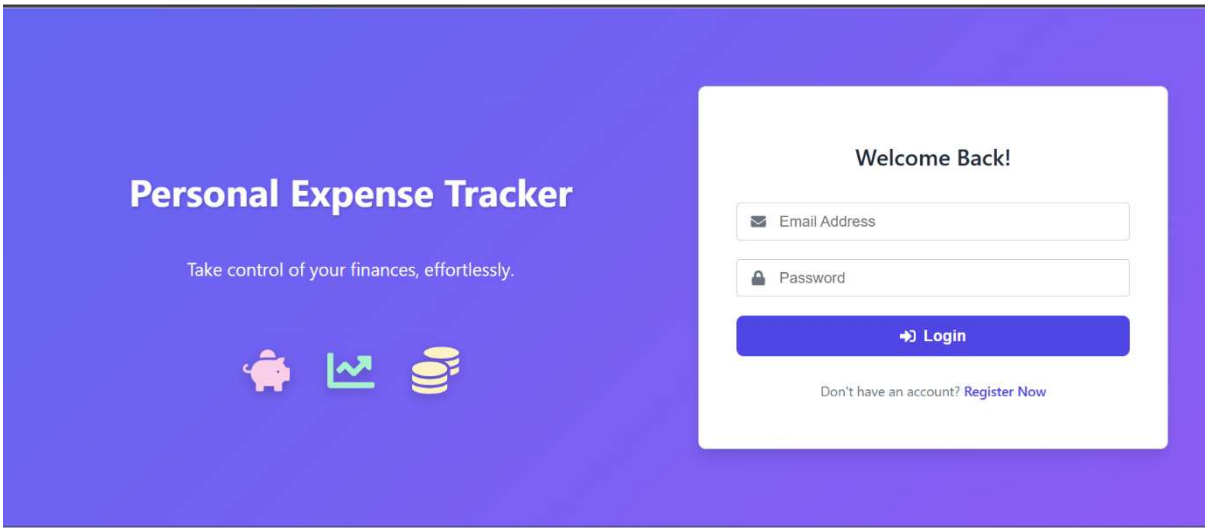
2. **Dashboard View**:



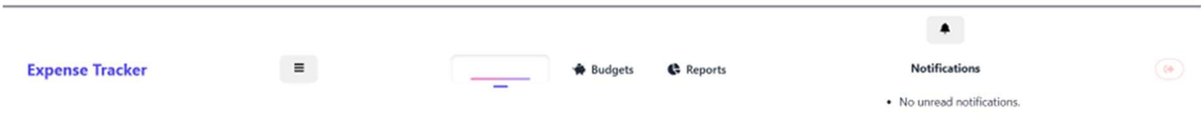
3. **Budgets Page**:



4. **Reports**:



5. **Notification Bell**:



8. ADVANTAGES & DISADVANTAGES

Advantages:

- User-friendly and intuitive interface
- Real-time expense and budget tracking
- Secure and scalable architecture
- Visual reports enhance decision-making
- Categorization supports various user types

Disadvantages:

- Lacks income tracking as of now
- Notifications are based on polling, not real-time WebSockets
- No mobile app for on-the-go access (yet)

9. CONCLUSION:

The *Personal Expense Tracker* successfully bridges the gap between awareness and action in financial management. By leveraging modern tools and intuitive design, the app encourages responsible spending and empowers users to take control of their finances. It combines form and function to deliver an impactful and scalable solution.

10. FUTURE SCOPE

- Real-Time Notifications: Switch from polling to WebSockets or Server-Sent Events.
- Income Tracking: Include income sources and financial inflow records.
- Recurring Transactions: Enable scheduled recurring expenses/incomes.
- User Roles: Multi-user accounts for families or small teams.
- Advanced Reporting: More visualization options, filtering, and CSV exports.
- Mobile Application: Create cross-platform mobile apps for convenience.
- Cloud File Uploads: Add receipt uploads using Cloudinary or AWS S3.
- Automated Testing: Introduce unit and integration tests for better QA.

11. APPENDIX

Source Code:

Frontend and Backend hosted at GitHub

➡ Link

Project Demo Video:

➡ Demo Link